

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

THE GEMINI PROTOCOLS · TOPICAL BOOK — LIVING VOLUME

THE ZERO G BOOK



FOR:

Everyone who works, plays, and builds with the
gravity switched off.

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THE GEMINI PROTOCOLS — THE ZERO G BOOK

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Commenced: 8 August 2026 — dictation seated verbatim, typos preserved, per file doctrine. *[IN PROGRESS — the design list has reached item 1 and continues by the author's dictation.]*

FOREWORD — REALITY OVER WISHFUL THINKING

I have contemplated reality over wishful / hopeful thinking regarding Zero G many times. Reality always comes back to gravity drive failure, if it fails and you did not build exactly this ship you die, not maybe, you die. no cooking, no food prep, no washing clothes, no magnetic floors, no fish tanks, no exercise regimes backpacks for spines etc etc etc. most notably when designing moon based equipment, it struck me that many similar issues will exist in low G moons and planets, there are certainly likely large planets greater G with Atmosphere as is likely low G with atmospheres.

From Mini Moons to the 15MW energy drive banks in low G on the moon, Zero G orbital engines that have passed a 3 AI physics vetting, ease of movement of multi-ton cargo on board, Zero G has huge benefits.

At the outset for those who have not read the orbital engine or the mini moon, i will quickly describe it.

Objects in motion is in physics perpetual motion, that is the law, any object in motion not affected by any other force will continue forever, an object traveling through an endless space void as an example. Mini moons and orbital engines and the cylinder engines on the Moon simply use that natural physics effect, on the moon, cylinders have axles suspended on repelling magnet bearings so they float like a maglev train; the vacuum does the rest.

An orbital engine is simply a magnet sphere suspended in a vacuum (space or a vacuum room) the sphere is encased in a shell, outside that is simply an outer shell covered in coil arrays, multiple or polar. covered and attached to the outer shell hull. with or without an axle. When the outer shell is spun up mechanically as simply as a carnival ride is with wheel sets then you have a generator. *[CORRECTED — 8 AUGUST 2026, THE AUTHOR'S RULING]:* the paragraph above describes the MINI-MOON arrangement (magnet inside, coil outside) — struck as the wrong build for the orbital engine, preserved per doctrine. The orbital engine is the REVERSE: **a coil sphere suspended in a vacuum (space or a vacuum room), the sphere encased in a shell; outside that, an outer shell carrying the tiled polarity MAGNET arrays, attached to the outer shell hull, with or without an axle. When the outer magnet shell is spun up mechanically — as simply as a carnival ride is with wheel sets — the stable coil center delivers the power: then you have a generator.** The coils sit at the stable center because that is where the power is

collected; the magnets rotate outside as tiled polarity shells with an outer field-filter layer, because giant neodymium spheres do not exist and could not be safely handled at the size required.

The variance is human monkey thought compared to mine, building any engine with say 10kw weight/power to a 10kw electrical motor is stupidity. A fly has no impact on the hull of a jumbo jet, the engines prorate are 500kw of weight power spin to 100kw engines. Lenz law is a small but real at standard weight momentum v electrical motors at equal scale, at 5-1 its not slowing these in years in a vacuum. My own preference for the vacuum rooms in zero G on ships is to still use a maglev axel, whilst small, object's are still subject to small movements over time in a vacuum box, ship impacts and inertia, basically you do not want a multi-ton spinning ball hitting the vacuum chamber wall.

Mini moons are magnet inside coil outside, orbital engines are the reverse. Giant neodymium magnets do not exist and if they did would be an impossibility to safety handle at the size required for an orbital engine as a sphere, tiled polarity shells are easily built, with an outer field filter layer. The other reason is the power generated must be collected, and from the stable center is easier than the rotating outer shell. On earth they could be built in vacuum chambers , but your output is -1G. the benefit gained in the 5-1 is lost to gravity. This is the reality, Zero G is a benefit and the safest build method.

These are the things that are designed for Zero G.

1.

THE LIST — THINGS DESIGNED FOR ZERO G (AND LOW G)

Compiled by the audit (Kimi) at the author's order, 8 August 2026 — "find them list them, you have a go going through the contents list". Every entry read out of the master's contents register (265 phases). Classification is the audit's pass: the author rules every strike and addition. Sections I-VII are zero-G shipboard; Section VIII is the low-G surface line the foreword calls for (Moon, planets, and the no-G rocks). SECOND SWEEP, 11 August 2026 (the author's order: "yeah do the sweep"): the register re-read from 266 to 274 plus Phase 2; nine new entries seated as Section IX, entries 68–76; entry 19 carries the Phase 265 renaming.

I. POWER & ENGINES — THE ZERO-G NATIVES

- **1. Phase 242 — The Mini-Moons.** Zero-powered orbiters; sun-charged phosphor shells; the spin battery priced.
- **2. Phase 249 — The Orbital Engine** *[RENAMED — the author's ruling, 8 August 2026; formerly The Wind-Up Moons]*. Objects in motion, honestly — the flywheel pod bank, two extraction designs flown head-to-head; coils at the stable center, tiled magnet polarity shells on the rotating outer shell — the reverse of the mini moons.

- **3. Phase 250 — The Magnetic Shaker.** The shaker torch upsized — gauss-primed zero-G momentum engine, pin-primed.
- **4. Phase 263 — The Work Light Stick.** The forever torch's zero-G child — 30mm x 300mm vacuum shaker stick, non-metal internals.
- **5. Phase 44 — Counter-Rotating Pinwheel Orbital Dynamos.** Wide-radius macro-scale kinetic in orbit with solar leverage.
- **6. Phase 101 — Frictionless High-Efficiency Induction Dynamo Tracks.** Wide-radius, axle-less stator tracks.
- **7. Phase 102 — Segmented External Slat Electromagnetic Maglev Rings.** Axle-less, center-free maglev ring — the orbital-engine line.
- **8. Phase 67 — Macro-Radial Low-Torque Dynamo Scaling.** The Tennis Ball Law; decadal pod lifecycle swaps.

II. LOW-G SURFACE POWER

- **9. Phase 264 — Moon Has Full Power.** The suburb-scale cylinder generator station — axles on repelling magnet bearings, vacuum does the rest.

III. THE GALLEY — COOKING & FOOD PREP WITHOUT GRAVITY

- **10. Phase 167 — The Universal Plunger Container.** The closed-loop 10 litre / 10 kilo supply standard.
- **11. Phase 168 — The Galley Transfer & Steam-Cook Protocol.** Screw-feed metering from sealed container to cook line.
- **12. Phase 169 — The Chewing Doctrine.** Mouth-feed dispensers, dining domes, portion cassettes — real texture in zero G.
- **13. Phase 170 — Magnetic Tableware & the Zero-G Wine Glass.** The Earth-feel doctrine at the table.
- **14. Phase 177 — The Birthday Cake Table.** Vacuum-downforce dining — the reverse air-hockey rule.
- **15. Phase 189 — The Space Microwave.** The clamped-plate rotator — zero-G hot food doctrine.
- **16. Phase 192 — The Zero-G Coffee Machine.** Fill by vacuum, stir by magnet — the no-open-cup doctrine.
- **17. Phase 198 — The Cake Oven — Horizontal Rotisserie & the Bubble Doctrine.** Locked-lid tin on the prong shaft, end over end; dual duty as the steak-and-roast oven (sitting 72).
- **18. Phase 222 — The Flour Cartridge.** Puncture-on-screw sealed powder dispensing; wet-first doctrine.

- **19. Phase 265 — The Figure-8 Fish Tank.** Counter-rotation aquaculture — the tank that doesn't rock.
[RENAME — THE AUTHOR'S RULING, 11 August 2026: "THE ZERO G AQUACULTURE TANK" — the phase now carries the book's own name.]

IV. WATER, WASHING & WASTE

- **20. Phase 64 — Centrifugal Trench-Gutter Fluid Stabilization.** Solves floating zero-G liquid chaos.
- **21. Phase 130 — Closed-Loop Dynamic Equilibrium Recirculation.** Dismantles zero-G bubble chaos in pressurized lines.
- **22. Phase 136 — Centrifugal Gutter-Feed Sports Bottle.** Drinking without gravity; rejects the space-bag.
- **23. Phase 137 — Centrifugal Waste Core & Polycarbonate Screw-Feed.** Waste without gravity plumbing; the vertical gutter, no gravity in the pipes.
- **24. Phase 144 — Dry Biowaste Pneumatic Mail Chute Loops.** Dry hygiene papers in compressed cardboard or gel.
- **25. Phase 146 — Barometric-Flux Space Shower.** The snub-nosed stall; showering without a drain.
- **26. Phase 190 — The Zero-G Dishwasher.** The hamburger with the lot — box in a bowl in a bowl.
- **27. Phase 191 — The Zero-G Washing Machine.** The Wringer Line — no spin, all pressure; the machine that never flew.
- **28. Phase 162 — Plunger-Lid Laundry & Greenhouse Drying Loop.** Top-load cylinder basket with flexible mesh lid.
- **29. Phase 145 — The Directional Fluid Momentum Directive.** Doctrine: running water requires no gravity.

V. THE BODY — EXERCISE, SPINE & BONE

- **30. Phase 99 — Biometric Lower-Limb Skeletal Compression.** Tricks the body's resource-optimization loop; bone load without gravity.
- **31. Phase 100 — Centrifugal Hydro-Kinetic Exercise Vault Pools.** Low-power rotating pool with retractable airtight lid.
- **32. Phase 171 — Downforce Cycles & the Exercise Velodrome.** Zero-G cycling via upward drone fans.
- **33. Phase 172 — The Gravity Hoodie.** Wearable axial downforce — the backpack for spines.
- **34. Phase 80 — Overhead Kinetic Harvesting Sky-Tracks & Inverted Exercise Grips.** Exercise that pays its own energy bill.

VI. LIVING & SPORT IN ZERO G

- **35. Phase 131 — Radial Carousel Sleep Pod Decks.** Sleep arranged as wheel spokes; phenotypic adaptation.
- **36. Phase 181 — The Zero-G Air Hockey Table.** The Birthday Cake, pump reversed.
- **37. Phase 182 — Hanging Badminton.** The shuttle that never falls.
- **38. Phase 183 — Battlestar & Death Star.** The 20-metre court; two weight classes, one war.
- **39. Phase 184 — Buddy Bowls.** The human jack — lawn bowls at human scale.
- **40. Phase 185 — The Iron Man Canister Course.** Cold-gas flight club.
- **41. Phase 186 — Curved-Track Rollerball.** Jets on, skates down.
- **42. Phase 193 — The Corridor Traffic Law.** Poles, rungs and walk lanes — the no-ceiling-jump rule.
- **43. Phase 243 — The Zero-G Snow Globe.** The toy that started it all — vacuum, no water.

VII. MOVING MASS — CARGO & CREW WITHOUT WEIGHT

- **44. Phase 54 — PEMS Pixelated Electromagnetic Maglev Surfaces.** The forklift-killers — multi-ton cargo moved by pixelated maglev.
- **45. Phase 41 — The Autonomous Omnidirectional Mule Tug Matrix.** Active linear tugs; no embedded rail propulsion.
- **46. Phase 20 — Extended High-Tensile Tethered Hangar Space-Anchor Tugs.** Heavy winch tethers for hangar mass.
- **47. Phase 39 — Spring-Lever Omni-Wheels & Silicone Sausage Locks.** Cargo pod running gear.
- **48. Phase 32 — Track-Interlock Boots & Induction-Safe Clearances.** Crew footing where there is no floor.
- **49. Phase 165 — Tether Gauntlets.** Wrist-mounted line throwers — a hold anywhere.

VIII. THE LOW-G SURFACE LINE — MOON, PLANETS & THE NO-G ROCKS

- **50. Phase 91/92/93 — The Suit Stack.** Carbon-film EVA layers, the Moon-suit mesh spacer strata, the dip coat and AION helmet.
- **51. Phase 262 — The Radiation Uniform.** The crossing uniform — the suit stack stripped for surface work.
- **52. Phase 206 — The Night Bank.** Passive lunar-night thermal machine — store the enemy; NASA Shortfall 1618.

- **53. Phase 257 — The Cavity-Brick Shelter.** The lunar build — two machines, foamy bricks, the enemy as armor.
- **54. Phase 258 — Water from Rocks.** The rocks-to-oxygen foundry.
- **55. Phase 259 — The Shield Tanks.** The shipment is the shelter — water wall-panels.
- **56. Phase 260 — The Shield Bricks.** 100mm water-filled borated-polyethylene wall lining, rods and holes, half-lap.
- **57. Phase 261 — Thermal Night Management — Vehicular.** The 30W frost pads, the 10C stat, the parking law.
- **58. Phase 84 — 5-Star Lunar-Agronomic Catering.** Dwarf banana flour, crystalline sugar — low-G agriculture (carries the fish-tank lineage seated now in 265).
- **59. Phase 33 — Zero-Atmosphere Outer-Rim Moon Depots.** Un-pressurized automated depots and in-flight module trades.
- **60. Phase 96 — Cryogenic Hull-Sinker Legs & Parachute-Free Ice Relief Drops.** Low-G landing without parachutes.
- **61. Phase 212 — No Gravity on a Space Rock.** Bag the rock, the Sun cooks, the cold side catches — asteroid water.
- **62. Phase 213 — Back to Archimedes.** The hot pin and the flared sleeve — contact cooking on the space rock.
- **63. Phase 219 — Shovel and Bubble Physics.** The free-face dig, design-for-fill ballast, the Mighty Quinn.
- **64. Phase 225 — The Roma Tomato Construction Rule.** Sequence is a free variable — the side-cut dig.
- **65. Phase 234 — The Mold Bank.** Lego-block houses, not bricks; the landing-bay vacuum works.
- **66. Phase 209 — The Spun Tank.** Thin liquid on a cold wall — rotating cryo storage.
- **67. Phase 210 — The Coffee Cup Rule.** Small cups, sealed lids, one open at a time — night-pod thermal rolls.

IX. THE SECOND SWEEP — 11 AUGUST 2026 (THE REGISTER RE-READ, 266–274 PLUS PHASE 2)

- **68. Phase 2 — The Self-Treading Endless Wheel.** Low-G surface mobility: radial hex tubes, spring-fed taper-locked rods — it cannot fall out, only wear out. The Mars-rover failure answered by geometry. (Natural home: Section VIII; commercial rubber-rod variant rides Earth Products.)

- **69. Phase 266 — The Tornado Box (The Bench Lightning Ball).** Nylon point-sphere inside the field wall, chopped-graphene bridge dust, the toroidal switch — zero-G bench power research. (Phase 267, The Narada Thread, merged in as Variant B.) (Natural home: Section I.)
- **70. Phase 268 — The Ten-Ring Stair.** Step up to 1G: ten graduated 6 m rings, the 3-second dwell on-ramp, the variable-g treadmill — the author's SANS fix. (Natural home: Section V.)
- **71. Phase 269 — The 5-1 Power Bank.** The orbital engine, scaled: 2 m/5 t ship shells at 500 rpm, the gyro-silent stack of eight; the 9 m moon machine — 4.76 kWh @ 500 rpm, 25 kWh @ the ceiling (Amendment 13 accepted spec). (Natural home: Sections I & II.)
- **72. Phase 270 — The Forever Tree.** The Archimedes scoop in a glow cone — ship variant: the cylinder for all events. (Natural home: Sections I & VI.)
- **73. Phase 271 — The Oxy-Defib Unit.** Wall-mounted emergency electrolysis oxygen — a litre of water is a person-day at rest. Zero-G-native emergency medicine. (Natural home: Section V.)
- **74. Phase 272 — The Memorial Torpedo.** Zero-power space burial — the spin-stable football pod on a catalogued thin vector. (Natural home: Section VIII, deep-space line.)
- **75. Phase 273 — The Telesurgery Suite [MEDICAL].** The two-pound surgical robot driven from anywhere — ship and county. (Natural home: Section V.)
- **76. Phase 274 — The Fifty-Year Feedstock.** Blocks, not spools — and the zero-G melt plant: cake-oven g-force crucibles, rotating-wall-screw collection, the not-airtight law. The printer that feeds the sweep. (Natural home: workshop/manufacture.)
- **77. Phase 1 — The Engine Chain.** The turbine's 200 m rotor knows no gravity sag — spin-up in free fall is precisely why it can be built stopped; the electrolyzers obey the ullage law (spin settles the water before the current flows, gas draws at the axis); babushka filling runs settled-spin transfers. Zero-g is not the chain's obstacle — it is its assembly shop.
- **78. Phase 290 — Gravity Drive Quarters & Gravitational Health Mandate.** The zero-G health problem answered outright with a machine: the 50 m G ring — 20 m wide, fed by the existing G-stairs, one fixed-colour wall and a step entrance so the rotation is never seen. Eight hours a day on deck for all aboard, by law; children live there to age nine; babies under two keep to the four-hour / two-hour windows until the skull closes. Pregnancy quarters, surgery with no floating blood, stable recoveries — and if the ring ever fails, nothing happens: the ship is zero-G anyway. (Natural home: The Medical Book, Section 10; the mandate law rides in Lex Archuria & Fleet Law.)

Audit note: seventy-eight entries. Items the author strikes leave the list marked STRUCK, never deleted; items he adds seat at the end with their own numbers. The foreword's own roll — cooking, food prep, washing clothes, magnetic floors, fish tanks, exercise regimes, backpacks for spines — each maps to a line above: cooking (10-17), food prep (10-12, 18), washing (26-28), floors-that-aren't (42, 44, 48), fish

tanks (19), exercise (30-34), the spine backpack (33). SECOND SWEEP (11 August 2026): entries now run to 77 — Section IX added; entry 19 carries the Phase 265 renaming to THE ZERO G AQUACULTURE TANK.